METRIC

DOD-R-23679F(SH)
8 August 1986
SUPERSEDING
DOD-R-23679E(SH)
16 November 1981
(See 6.6)

MILITARY SPECIFICATION

REAGENTS, STANDARD, WATER TEST CHEMICALS (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers reagents for use in checking the quality of water testing chemicals and equipments used in boiler powered ships.

1.2 Classification. Standard reagents shall be of the following classes, as specified (see 6.3).

Class e - Standard hardness solution, 0.20 equivalent per million (e/m hardness).
Class i - Standard conductivity solution, 1400 microsiemens per centimeter (us/cm).
Class k - Standard chloride solution, 2.00 e/m chloride.
Class m - pH standard, buffer capsules pH - 10.00 ± 0.05 at 25 degrees Celsius (°C).
Class n - pH standard, buffer capsules pH - 7.00 ± 0.05 at 25°C.
Class p - Standard methyl purple alkalinity standard 6.0 e/m.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-W-15000 - Water-Testing Chemicals, Boiler, Shipboard Use.

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets Preparation and the Submission of.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (OSHA)

Code of Federal Regulations (CFR)

CFR 29, Part 1910.1200 - Hazard Communication Standard.

DEPARTMENT OF TRANSPORTATION (DOT)

Code of Federal Regulations (CFR)

CFR 49, Parts 170-179 - Hazardous Material Regulations.

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(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN CHEMICAL SOCIETY (ACS)
Specifications - Reagent Chemicals.

(Application for copies should be addressed to Applied Publications, American Chemical Society, 1155 16th Street, NW, Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 512 - Standard Test Methods for Chloride Ion in Water. (DoD adopted)
D 1125 - Standard Test Methods for Electrical Conductivity and Resistivity of Water. (DoD adopted)
D 1126 - Standard Test Method for Hardness in Water.
D 1193 - Standard Specification for Reagent Water. (DoD adopted)
D 1293 - Standard Test Methods for pH of Water. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard reagents. All reagents shall conform to specifications of the ACS. Manufacturer and lot number for ACS grade reagents shall be provided to the Government laboratory with samples for quality conformance testing as specified in 4.2.3.

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3.1.1 Class e - standard hardness solution, 0.20 e/m hardness.

3.1.1.1 Composition. Standard hardness solution shall contain 0.0111 grams of reagent grade anhydrous calcium chloride per liter (L) of type II reagent grade water solution. When tested in accordance with 4.4.1, the total hardness in milliequivalents per liter (meq/L) shall be 0.20 ± 0.03 meq/L.

3.1.2 Class i - standard conductivity solution, 1400 µs/cm.

3.1.2.1 Composition. Standard conductivity solution shall contain 0.7440 grams of reagent grade anhydrous potassium chloride per L of type II reagent grade water solution. When tested in accordance with 4.4.2, the conductivity value of the solution shall be 1400 ± 25 µs/cm.

3.1.3 Class k - standard chloride solution, 2.00 e/m chloride.

3.1.3.1 Composition. Standard chloride solution shall contain 0.1170 grams of reagent grade anhydrous sodium chloride per L of type II reagent grade water solution. When tested in accordance with 4.4.3, the chloride ion concentration in milligrams per L (mg/L) shall be 70.91 ± 1.40 mg/L.

3.1.4 Class m - pH standard, pH - 10.00 + 0.05 at 25°C.

3.1.4.1 The pH standard shall be a stable buffer powder in capsule form. When the contents of one capsule are dissolved in 100 milliliters (mL) of type II reagent grade water at 25°C, the resultant solution shall be free of turbidity or undissolved particles. When tested in accordance with 4.4.4, this solution shall yield a pH of 10.00 ± 0.05.

3.1.5 Class n - pH standard, pH 7.00 + 0.05 at 25°C.

3.1.5.1 The pH standard shall be a stable buffer powder in capsule form. When the contents of one capsule are dissolved in 100 mL of type II reagent grade water at 25°C, the resultant solution shall be free of turbidity or undissolved particles. When tested in accordance with 4.4.5, this solution shall yield a pH of 7.00 ± 0.05.

3.1.6 Class p - standard methyl purple alkalinity (6.0 e/m).

3.1.6.1 Composition. The standard shall consist of 0.318 grams of ACS reagent grade sodium carbonate per L of distilled or demineralized water solution. The sodium carbonate shall conform to the Specifications for Reagent Chemicals of the ACS. The water shall be in accordance with ASTM D 1193, type III. The sodium carbonate shall be dried in an oven at 250°C for 4 hours then cooled in a desiccator prior to preparing the standard. When tested in accordance with 4.4.6, the alkalinity in e/m should not be more than 6.2 nor less than 5.8.

3.2 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200, Hazard Communication. When FED-STD-313 is at variance with the CFR, 29 CFR 1910.1200 shall take precedence, modify and supplement FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.2 Quality conformance inspection.

4.2.1 Lot. For purposes of sampling, a lot shall consist of all solutions or chemicals from one batch of a single class.

4.2.2 Sampling for examination of filled containers. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at inspection level I, and acceptable quality level (AQL) of 2.5 percent defective to verify compliance with all requirements of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.2.3 Sampling for quality conformance tests. Unless a different laboratory is specified (see 6.2.1), 60 bottles shall be selected from each lot of material and forwarded to the David W. Taylor Naval Ship Research and Development Center, Annapolis Laboratory, Annapolis, MD 21402 for the tests specified in 4.4 and 4.5. A copy of the test and evaluation report for each lot, noting contract number, national stock number, lot/batch number and date of manufacture, shall be included, as specified (see 6.2.2).

4.2.3.1 Classes e, i, k, and p. The results of the tests shall be employed to estimate the percent nonconforming in the lot by computations in accordance with MIL-STD-414, example B-3. The maximum allowable percent outside the specification limits shall be 4.0 percent for both limits combined.

4.2.3.2 Classes m and n. One capsule from each bottle shall be dissolved in 100 mL of distilled water and tested as specified in 4.4.4 or 4.4.5. If any buffer fails this test, this shall be cause for rejection of the lot.

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4.2.3.3 If the computations as specified in 4.2.3.1 show that more than 4.0 percent of the lot is outside the specification limits for any class, this shall be cause for rejection of the entire lot.

4.2.3.4 Delivery of the material shall be withheld pending notification of the results of tests.

4.3 Examinations.

4.3.1 Examination of filled containers. Each sample filled container selected in accordance with 4.2.2 shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Any container in the sample, having one or more defects, or under required fill, shall be cause for rejection, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan in accordance with MIL-STD-105, this shall be cause for rejection of the lot represented by the sample.

4.4 Tests. Each of the samples selected and forwarded in accordance with 4.2.3 shall be subjected to the applicable tests specified herein. All tests shall be conducted at room temperature.

4.4.1 Class e - standard hardness solution, 0.20 e/m hardness (see 4.4.7). Measure the total hardness concentration in accordance with ASTM D 1126, 9.2, low hardness method. Record value. The calcium solution standard used for standardizing the disodium ethylenediamine tetraacetate solution shall be prepared using Standard Reference Material calcium carbonate from the National Bureau of Standards.

4.4.2 Class i - standard conductivity solution, 1400 us/cm (see 4.4.7). Determine the cell constant of the conductivity cell which will be used for testing the standard conductivity solution in accordance with ASTM D 1125, method A. The potassium chloride reference solution shall be prepared using Standard Reference Material potassium chloride from the National Bureau of Standards. Measure electrical conductivity of the standard conductivity solution in accordance with ASTM D 1125, method A. Record value.

4.4.3 Class k - standard chloride solution, 2.00 e/m chloride (see 4.4.7). Measure the chloride ion concentration in accordance with ASTM D 512, method A (mercurimetric, titration). Record value. The sodium chloride reference solution used for standardizing the mercuric nitrate solution shall be prepared using Standard Reference Material sodium chloride from the National Bureau of Standards.

4.4.4 Class m - pH standard, pH 10.00 + 0.05 at 25°C. Dissolve the contents of one capsule in 100 mL of type II reagent grade water at 25°C, and determine pH in accordance with ASTM D 1293, method A. Record value.

4.4.5 Class n - pH standard, pH 7.00 + 0.05 at 25°C. Dissolve the contents of one capsule in 100 mL of type II reagent grade water at 25°C, and determine pH in accordance with ASTM D 1293, method A. Record value.

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4.4.6 Class p - standard methyl purple alkalinity (6.0 e/m) (see 4.4.7).

4.4.6.1 The standard shall contain not more than 6.2 e/m nor less than 5.8 e/m alkalinity when tested by the following procedure.

(a) Pipet 50 mL of the standard into a clean casserole.
(b) Add 2-3 drops of methyl purple indicator. The methyl purple indicator shall be in accordance with MIL-W-15000.
(c) Titrate carefully with standardized 0.05N nitric acid until a purple endpoint is obtained.
(d) Calculate alkalinity as follows:

alkalinity, e/m = A X B/0.0500

Where:

A = mL of nitric acid
B = normality of the nitric acid obtained during standardization

4.4.6.2 The nitric acid shall be standardized against 25 mL of 0.05N sodium carbonate solution prepared from Standard Reference Material from the National Bureau of Standards using methyl purple indicator. Once the endpoint is obtained, the buret reading shall be recorded. The sodium carbonate solution shall not be boiled. Titration shall be continued. The normality of the nitric acid shall be calculated as the average of three determinations.

4.4.7 Classes e, i, k, and p. Prior to performing each test, the contents of each bottle shall be transferred to a TC graduated cylinder, having a cylinder volume sufficient to contain the bottle contents. The volume of standard shall be measured. The value shall be recorded.

4.5 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2.1).

5.1.1 Levels A and C.

5.1.1.1 Compatibility of container. The container, closure, lining, or space fillers shall not interact physically or chemically with the contents of the container so as to be altered or to alter the strength, quality or purity of the container contents for at least 2 years from the date of manufacture.

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5.1.1.2 Containers, classes e, i, k, and p. No less than 75 mL of standard chloride solution, class k, shall be furnished in 89 mL (3 ounce) bottles. No less than 150 mL of standard hardness solution, class e, shall be furnished in 177 mL (6 ounce) bottles. No less than 100 mL of standard methyl purple alkalinity (6.0 e/m), class p, shall be furnished in 118 mL (4 ounce) bottles. No less than 250 mL of standard conductivity solution, class i, shall be furnished in 266 mL (9 ounce) bottles. All bottles shall be in accordance with group A, class 2, style 1 of PPP-C-186. Closures shall be polyethylene or polypropylene, continuous thread, screw type caps. Caps shall be secured with a cellulose band. The band shall extend down on the neck of the bottle and over the edge of the cap.

5.1.1.2.1 Six bottles of classes e, k, p, and i shall be packaged into a paperboard partitioned box. The box shall be closed with pressure-sensitive tape.

5.1.1.3 Containers, classes m and n. The pH standard, classes m and n, shall be furnished in quantities of 10 capsules. Capsules containing pH 7 buffer shall be yellow in color. Capsules containing pH 10 buffer shall be blue in color. Each capsule shall be individually packaged in heat-sealed foil. Ten capsules shall be packaged in a glass or plastic vial. Vials shall be in accordance with PPP-C-186, group A, class 1 - glass or class 2 - plastic, style 6. Grade of light penetration for group A containers and the glass or plastic type shall be at the contractor's option. Closures for group A containers shall conform to closure A, I, or J at the contractor's option.

5.1.1.3.1 Five vials of a single class shall be packaged in a plastic tray which incorporates individual pockets or cells for holding each vial in a snug-fitting, horizontal (nonreversible) position. Each tray shall then be inserted into a tuck-end paperboard folding type box. Box closures shall be secured with pressure-sensitive tape.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1).

5.2.1 Level A. Twelve boxes of bottles or vials preserved as specified in 5.1, shall be packed in nailed wood or cleated plywood boxes conforming to PPP-B-621, class 2 or PPP-B-601 overseas type respectively, with container selection at the option of the contractor. Unless otherwise specified (see 6.2.1), boxes shall have caseliners conforming to MIL-L-10547. Boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of each shipping container shall not exceed 91 kilograms (kg) (200 pounds).

5.2.2 Level B. Twelve boxes of bottles or vials preserved as specified in 5.1, shall be packed in fiberboard boxes conforming to the type CF, class weather-resistant in accordance with PPP-B-636. The variety, grade, and style shall be at the contractor's option. Box closure shall be in accordance with method V of the appendix to the box specification.

5.2.3 Level C. Twelve boxes of bottles or vials preserved as specified in 5.1, shall be packed in fiberboard boxes in accordance with type CF, class domestic in accordance with PPP-B-636. The variety, grade, and style shall be at the contractor's option. Box closure shall be in accordance with method I of the appendix to the box specification.

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5.3 Marking. In addition to any special marking required (see 6.2.1) and herein, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129, CFR 29, Part 1910.1200, CFR 49, Parts 170-179 and as follows.

5.3.1 Interior packages (bottles, vials, and boxes).

5.3.1.1 Bottles and vials. In addition to the identification markings required in 5.3, each bottle and vial shall be marked with the contents' manufacturing and expiration dates (for example, manufactured 4/76 expires 4/78), manufacturer's or contractor's name, address, contract number, and batch or lot number. In addition, each bottle and vial, as applicable, containing the following contents, shall be provided the following marking:

Standard hardness solution. "Standard Hardness Solution 0.20 e/m hardness."

pH Standard, pH 10.00. "pH Standard - The contents of each capsule when dissolved in 100 mL of distilled water yield a solution of pH 10.00 at 25°C."

Conductivity standard. "Standard Conductivity Solution - 1400 µs/cm."

pH Standard, pH 7.00. "pH Standard - The contents of each capsule when dissolved in 100 mL of distilled water yield a solution of pH 7.00 at 25°C."

Standard chloride solution. "Standard Chloride Solution - 2.00 e/m chloride."

Standard methyl purple alkalinity. "Standard Methyl Purple Alkalinity, 6.0 e/m."

5.3.1.2 Boxes. Each box of bottles or vials shall be identified as specified in 5.3.1.1.

5.3.2 Exterior containers. Each shipping container, in addition to the identification and markings as specified in 5.3, shall be marked as specified in 5.3.3.

5.3.3 Packages and shipping containers. In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129, CFR 29, Part 1910.1200, CFR 49, Parts 170-179 and shall include the following:

(a) Name and class of item.
(b) National stock number.
(c) Manufacturer's or contractor's name and address.
(d) Batch or lot number.
(e) Contract and order number.
(f) Quantity.
(g) Date of manufacture, month, and year.
(h) Expiration date of 24 months from date of manufacture - for example, manufactured 4/76 expires 4/78.

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6. NOTES

6.1 Intended use. The reagents covered by this specification are intended to be used as a means for checking on the quality of chemicals and equipment used in testing shipboard boiler water and feedwater.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class of chemical required (see 1.2).
(c) Laboratory desired for Government testing (see 4.2.3).
(d) Levels of preservation and of packing required (see 5.1 and 5.2).
(e) When caseliners are not required (see 5.2.1).
(f) Special markings required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1.2	Inspection system program plan	DI-R-4803	----
4.2.3	Test and evaluation report	DI-T-4909	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., Vol. I, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 Standard solutions cross-reference. The standard solution classes in this specification and the corresponding classes in DOD-R-23679E(SH) are listed below:

TABLE I. Standard solution classes cross-reference.

DOD-R-23679F(SH)	DOD-R-23679E(SH)
-----	-----
-----	-----
class e	class e
-----	-----
-----	-----
-----	class h
class i	class i
-----	class j
class k	class k
-----	class *l*
class m	-----
class n	-----
class p	-----

6.4 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313.

6.5 Subject term (key word) listing.

Chemicals
Chloride
Conductivity
Hardness
Methyl purple
pH
Reagents
Water test

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6810-NB55)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorizatio to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-R-23679F(SH)	REAGENTS, STANDARD, WATER TEST CHEMICALS (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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